## Chapter 2

**4.**

**a.** The $2^4 = 16$ possible outcomes have been numbered here for later reference.

| Outcome | Home Mortgage Number 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | *F* | *F* | *F* | *F* |
| 2 | *F* | *F* | *F* | *V* |
| 3 | *F* | *F* | *V* | *F* |
| 4 | *F* | *F* | *V* | *V* |
| 5 | *F* | *V* | *F* | *F* |
| 6 | *F* | *V* | *F* | *V* |
| 7 | *F* | *V* | *V* | *F* |
| 8 | *F* | *V* | *V* | *V* |
| 9 | *V* | *F* | *F* | *F* |
| 10 | *V* | *F* | *F* | *V* |
| 11 | *V* | *F* | *V* | *F* |
| 12 | *V* | *F* | *V* | *V* |
| 13 | *V* | *V* | *F* | *F* |
| 14 | *V* | *V* | *F* | *V* |
| 15 | *V* | *V* | *V* | *F* |
| 16 | *V* | *V* | *V* | *V* |

**b.** Outcome numbers 2, 3, 5, 9 above.

**c.** Outcome numbers 1, 16 above.

**d.** Outcome numbers 1, 2, 3, 5, 9 above.

**e.** In words, the union of (c) and (d) is the event that either all of the mortgages are variable, or that at most one of them is variable-rate: outcomes 1, 2, 3, 5, 9, 16. The intersection of (c) and (d) is the event that all of the mortgages are fixed-rate: outcome 1.

**f.** The union of (b) and (c) is the event that either exactly three are fixed, or that all four are the same: outcomes 1, 2, 3, 5, 9, 16. The intersection of (b) and (c) is the event that exactly three are fixed and all four are the same type. This cannot happen (the events have no outcomes in common), so the intersection of (b) and (c) is $\varnothing$.

**6.**

**a.** $S = \{123, 124, 125, 213, 214, 215, 13, 14, 15, 23, 24, 25, 3, 4, 5\}$.

**b.** $A = \{3, 4, 5\}$.

**c.** $B = \{125, 215, 15, 25, 5\}$.

**d.** $C = \{23, 24, 25, 3, 4, 5\}$.

**12.**

**a.** No, this is not possible. Since event $A \cap B$ is contained within event $B$, it must be the case that $P(A \cap B) \leq P(B)$. However, $.5 > .4$.

**b.** By the addition rule, $P(A \cup B) = \cancel{.5}\ .6 + .4 - .3 = \cancel{.6}\ .7$.

**c.** $P(\text{neither } A \text{ nor } B) = P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) = 1 - \cancel{.6}\ .7 = \cancel{.4}\ .3$.

**d.** The event of interest is $A \cap B'$; from a Venn diagram, we see $P(A \cap B') = P(A) - P(A \cap B) = \cancel{.3}\ .6 - .3 = \cancel{.1}\ .3$

**e.** From a Venn diagram, we see that the probability of interest is $P(\text{exactly one}) = P(\text{at least one}) - P(\text{both}) = P(A \cup B) - P(A \cap B) = \cancel{.6}\ .7 - .3 = \cancel{.3}\ .4$

**14.** Let $A$ = an adult consumes coffee and $B$ = an adult consumes carbonated soda. We're told that $P(A) = .55$, $P(B) = .45$, and $P(A \cup B) = .70$.

**a.** The addition rule says $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, so $.70 = .55 + .45 - P(A \cap B)$ or $P(A \cap B) = .55 + .45 - .70 = .30$.

**b.** There are two ways to read this question. We can read "does not (consume at least one)," which means the adult consumes neither beverage. The probability is then $P(\text{neither } A \text{ nor } B) = P(A' \cap B') = 1 - P(A \cup B) = 1 - .70 = .30$.

The other reading, and this is presumably the intent, is "there is at least one beverage the adult does not consume, i.e. $A' \cup B'$. The probability is $P(A' \cup B') = 1 - P(A \cap B) = 1 - .30$ from **a** = .70. (It's just a coincidence this equals $P(A \cup B)$.)

Both of these approaches use *deMorgan's laws*, which say that $P(A' \cap B') = 1 - P(A \cup B)$ and $P(A' \cup B') = 1 - P(A \cap B)$.

**18.** The only reason we'd need at least two selections to find a \$10 bill is if the <u>first</u> selection was <u>not</u> a \$10 bill bulb. There are 4 + 6 = 10 non-\$10 bills out of 5 + 4 + 6 = 15 bills in the wallet, so the probability of this event is simply 10/15, or 2/3.

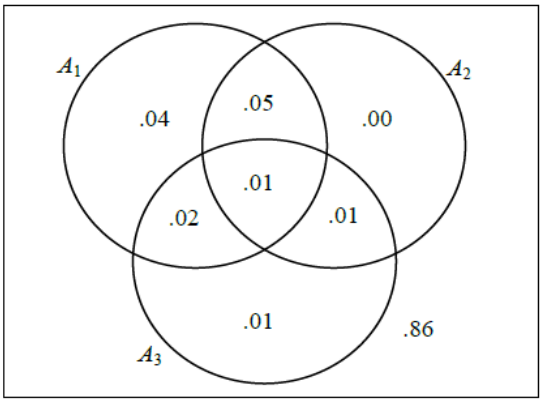
**26.** These questions can be solved algebraically, or with the Venn diagram below.

**a.** $P(A_1') = 1 - P(A_1) = 1 - .12 = .88$.

**b.** The addition rule says $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Solving for the intersection ("and") probability, you get $P(A_1 \cap A_2) = P(A_1) + P(A_2) - P(A_1 \cup A_2) = .12 + .07 - .13 = .06$.

**c.** A Venn diagram shows that $P(A \cap B') = P(A) - P(A \cap B)$. Applying that here with $A = A_1 \cap A_2$ and $B = A_3$, you get $P([A_1 \cap A_2] \cap A_3') = P(A_1 \cap A_2) - P(A_1 \cap A_2 \cap A_3) = .06 - .01 = .05$.

**d.** The event "at most two defects" is the complement of "all three defects," so the answer is just $1 - P(A_1 \cap A_2 \cap A_3) = 1 - .01 = .99$.




**30.**

**a.** Because order is important, we'll use $P_{3,8} = (8)(7)(6) = 336$.

**b.** Order doesn't matter here, so we use $\binom{30}{6} = 593{,}775$.

**c.** The number of ways to choose 2 zinfandels from the 8 available is $\binom{8}{2}$. Similarly, the number of ways to choose the merlots and cabernets are $\binom{10}{2}$ and $\binom{12}{2}$, respectively. Hence, the total number of options (using the Fundamental Counting Principle) equals $\binom{8}{2}\binom{10}{2}\binom{12}{2} = (28)(45)(66) = 83{,}160$.

**d.** The numerator comes from part **c** and the denominator from part **b**: $\frac{83{,}160}{593{,}775} = .140$.

**e.** We use the same denominator as in part **d**. The number of ways to choose all zinfandel is $\binom{8}{6}$, with similar answers for all merlot and all cabernet. Since these are disjoint events, $P(\text{all same}) = P(\text{all zin}) + P(\text{all merlot}) + P(\text{all cab}) = \dfrac{\binom{8}{6} + \binom{10}{6} + \binom{12}{6}}{\binom{30}{6}} = \dfrac{1162}{593{,}775} = .002$.

**34.**

**a.** Since order doesn't matter, the number of ways to randomly select 5 keyboards from the 25 available is $\binom{25}{5} = 53{,}130$.

**b.** Sample in two stages. First, there are 6 keyboards with an electrical defect, so the number of ways to select exactly 2 of them is $\binom{6}{2}$. Next, the remaining 5 – 2 = 3 keyboards in the sample must have mechanical defects; as there are 19 such keyboards, the number of ways to randomly select 3 is $\binom{19}{3}$. So, the number of ways to achieve both of these in the sample of 5 is the product of these two counting numbers: $\binom{6}{2}\binom{19}{3} = (15)(969) = 14{,}535$.

**c.** Following the analogy from **b**, the number of samples with exactly 4 mechanical defects is $\binom{19}{4}\binom{6}{1}$, and the number with exactly 5 mechanical defects is $\binom{19}{5}\binom{6}{0}$. So, the number of samples with <u>at least</u> 4 mechanical defects is $\binom{19}{4}\binom{6}{1} + \binom{19}{5}\binom{6}{0}$, and the probability of this event is

$$\frac{\binom{19}{4}\binom{6}{1}+\binom{19}{5}\binom{6}{0}}{\binom{25}{5}} = \frac{34{,}884}{53{,}130} = .657.$$ (The denominator comes from **a**.)

**36.** There are $\binom{5}{2} = 10$ possible ways to select the positions for *B*'s votes: *BBAAA, BABAA, BAABA, BAAAB, ABBAA, ABABA, ABAAB, AABBA, AABAB*, and *AAABB*. Only the last two have *A* ahead of *B* throughout the vote count. Since the outcomes are equally likely, the desired probability is 2/10 = .20.

**50.**

**a.** $P(\mathbf{M} \cap \mathbf{LS} \cap \mathbf{PR}) = .05$, directly from the table of probabilities.

**b.** $P(\mathbf{M} \cap \mathbf{Pr}) = P(\mathbf{M} \cap \mathbf{LS} \cap \mathbf{PR}) + P(\mathbf{M} \cap \mathbf{SS} \cap \mathbf{PR}) = .05 + .07 = .12$.

**c.** $P(\mathbf{SS})$ = sum of 9 probabilities in the **SS** table = .56. $P(\mathbf{LS}) = 1 - .56 = .44$.

**d.** From the two tables, $P(\mathbf{M}) = .08 + .07 + .12 + .10 + .05 + .07 = .49$. $P(\mathbf{Pr}) = .02 + .07 + .07 + .02 + .05 + .02 = .25$.

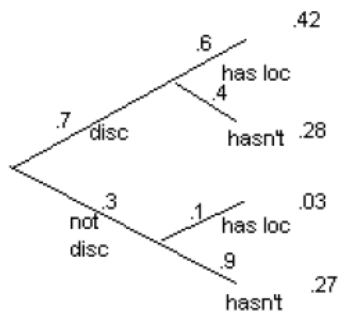
**e.** $P(\mathbf{M}|\mathbf{SS} \cap \mathbf{Pl}) = \dfrac{P(\mathbf{M} \cap \mathbf{SS} \cap \mathbf{Pl})}{P(\mathbf{SS} \cap \mathbf{Pl})} = \dfrac{.08}{.04+.08+.03} = .533$.

**f.** $P(\mathbf{SS}|\mathbf{M} \cap \mathbf{Pl}) = \dfrac{P(\mathbf{SS} \cap \mathbf{M} \cap \mathbf{Pl})}{P(\mathbf{M} \cap \mathbf{Pl})} = \dfrac{.08}{.08+.10} = .444$. $P(\mathbf{LS}|\mathbf{M} \cap \mathbf{Pl}) = 1 - P(\mathbf{SS}|\mathbf{M} \cap \mathbf{Pl}) = 1 - .444 = .556$.

**56.** $$P(A \mid B) + P(A' \mid B) = \frac{P(A \cap B)}{P(B)} + \frac{P(A' \cap B)}{P(B)} = \frac{P(A \cap B) + P(A' \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

**58.** $$P(A \cup B \mid C) = \frac{P[(A \cup B) \cap C)}{P(C)} = \frac{P[(A \cap C) \cup (B \cap C)]}{P(C)} = \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} = P(A \mid C) + P(B \mid C) - P(A \cap B \mid C)$$

**60.** The tree diagram below shows the probability for the four disjoint options; e.g., $P$(the flight is discovered and has a locator) = $P$(discovered)$P$(locator | discovered) = (.7)(.6) = .42.




**a.** $P(\text{not discovered} \mid \text{has locator}) = \dfrac{P(\text{not discovered} \cap \text{has locator})}{P(\text{has locator})} = \dfrac{.03}{.03 + .42} = .067$.

**b.** $P(\text{discovered} \mid \text{no locator}) = \dfrac{P(\text{discovered} \cap \text{no locator})}{P(\text{no locator})} = \dfrac{.28}{.55} = .509$.

**74.** Using subscripts to differentiate between the selected individuals,
$P(O_1 \cap O_2) = P(O_1)P(O_2) = (.45)(.45) = .2025$.
$P$(two individuals match) = $P(A_1 \cap A_2) + P(B_1 \cap B_2) + P(AB_1 \cap AB_2) + P(O_1 \cap O_2) = .40^2 + .11^2 + .04^2 + .45^2 = .3762$.

**78.** $P$(at least one opens) = 1 – $P$(none open) = $1 - (.04)^5 = .999999897$.
$P$(at least one fails to open) = 1 – $P$(all open) = $1 - (.96)^5 = .1846$.